

TECHNOLOGY

With a \$60-billion information

technology industry, India is certainly on the world's technology radar. As Kapil Sibal, minister of science and technology, predicts, "India is poised to become a powerhouse of innovation in the years to come." We are getting there, slowly. India spends about \$45 billion on research and development, compared to China, which spends \$216 billion. In 2007-08, a record 15,262 patents were granted in India, compared to 7,500 the previous year. And when citizens set out to make a difference, it signals a revolution. Take Joseph Appachan who, with a new kind of device, made life easy for coconut tree climbers in Kerala, or Mujeeb Khan, who helps make the handicapped self-reliant by designing cars especially for them. Or H. Harish Hande, whose company distributes solar-powered lights to low-income groups. It could be a natural water cooler or a new, improved cotton stripper. What makes things work better makes life more liveable.

■ By M.G. Radhakrishnan

Kerala literally means the land of the coconut palm and there isn't a patch in the picturesque state without this ubiquitous tree. Though the coconut or *nalikeram* in Malayalam has always been a part of the state's culture, cuisine and commerce, Kerala is facing a major problem now. There is an acute shortage of

traditional coconut climbers and other manual labourers related to this age-old profession. The situation has become so grave in a state with about 16 crore palm trees that the authorities have decided to start a free training school for interested candidates and award diplomas to them. A special scheme is already running in a school in Malappuram, where women are trained as tree climbers, a profession which was once considered a male bastion.

But with all initiatives taken up by the Government failing to improve the situation, M.J. Joseph Appachan's innovative talent has filled an enormous vacuum. The 50-year-old small scale farmer and a high school dropout with the brain of a techie has invented a coconut tree-climbing device which can be used by anyone to climb a tree to pluck nuts, tap the basic ingredient of toddy, the indigenous liquor, or spray insecticides. The simple and safe device consists of two steel foot pedals connected by wires and steel pipes with which one can climb a 40-m palm in just two to three minutes. This time equals half of what one usually needs to reach the tree top. Appachan's user-friendly innovation can be credited to his father, also called Joseph, who developed the original device way back in 1984 at a time when nobody thought that

the state would ever face such a crisis. "The credit goes to my father. I have only modified his prototype," says Appachan, sitting in front of his St Mary's Engineering Workshop in an obscure hilly village in Kannur.

Each village, which used to have 100 climbers in the past, is now left with only two to three. This has kicked up the demand for the few available climbers who have raised their charges from Rs 5 per tree to Rs 100. As a result, they are now out of bounds for many middle-class households who have just four to five palms in their homestead. "Ten years ago, our family thandan (a caste denoting climbers) used to visit our house at least twice a month to pluck coconuts. Today, I have to call the son of the old thandan, to come and pluck the coconuts. I have to ferry him back and forth in my car," says M.K. Nair, a retired bureaucrat in Thiruvananthapuram. The younger generations of such families have opted for more "respectable" professions, and around 20 lakh have migrated to the Gulf so far.

Appachan is a devout Christian and a hard working farmer. His only formal training was a metal smith's course when he was young. "I learnt to bend steel and do some welding there," he says. His wife, Sali, helps him in his farm and works

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We tell the Americans you may be able to go to the moon but you will still need Appachan's climbing device to go up a tree.

ANIL GUPTA, Vice-chairman, NIF

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